When Senses Disagree: Conflict-Aware Arbitration for Multimodal Reasoning

Otto Xin
Department of Communication Studies
Northwestern University
Evanston, IL 60208
haohangxin@u.northwestern.edu

Nicholas Ornstein
Department of Communication Studies
Northwestern University
Evanston, IL 60208
nickornstein@u.northwestern.edu

Abstract

Multimodal large language models (MLLMs) are brittle when image and text inputs disagree: instead of recognizing conflict, they often answer confidently with information from the wrong modality. We frame this as an arbitration problem, where a model must determine whether each modality is informative and whether informative modalities agree. We propose **CARM** (Conflict-Aware Reasoning Module), a lightweight module for frozen MLLMs that predicts modality informativeness, cross-modal relation, and a final action in a staged cascade. We evaluate CARM on Conflict Suite, a 44,948-example benchmark built from VQAv2/COCO under a five-category conflict protocol. On the canonical full split (31,463 train / 6,743 validation / 6,742 test) with Qwen2.5-VL-7B, CARM reaches 66.2% task success on the test set, improving over the strongest non-learned baseline from 48.9% to 66.2%. A full-data ablation shows that hidden-state features account for most of the gain, while the cascade adds a smaller further improvement from 65.1% to 66.2% over a flat hidden-state control. These results suggest that multimodal conflict handling is better modeled as explicit arbitration than as generic abstention or confidence estimation.

1 Introduction

Multimodal large language models (MLLMs) are attractive because they can combine images and language in a single reasoning pipeline, but they become unreliable when those modalities disagree. In realistic deployments, such disagreements are common: captions can be stale, retrieved text can describe the wrong image, and one modality can be ambiguous while the other is decisive. In these cases, current MLLMs often produce fluent but evidentially ungrounded answers instead of acknowledging that the modalities conflict or that one of them should be ignored (Hua et al., 2025; Popordanoska et al., 2025; Zhang et al., 2025c; Deng et al., 2025; Yan et al., 2025).

The key difficulty is that conflict handling is a structured decision problem, not just a generic uncertainty problem. A system must determine whether the image is informative for the question, whether the text is informative, and, if both are informative, whether the two modalities agree. Existing mitigations typically operate at the wrong level of abstraction. Prompting-based verification and generic abstention heuristics can reduce some errors, but they do not explicitly model the arbitration structure of the problem itself (Zhang et al., 2025a; Wen et al., 2025). This leads to two familiar mistakes: over-commitment when both modalities are confident but contradictory, and over-abstention when only one modality is useful.

We address this gap with CARM (Conflict-Aware Reasoning Module), a lightweight module that attaches to a frozen MLLM backbone and makes the arbitration structure explicit. Our central empirical observation is that, for Qwen2.5-VL-7B, last-layer hidden states from the full multimodal pass provide a much stronger arbitration signal than output distributions alone. CARM therefore combines two unimodal probes with hidden-state features from the joint pass, then predicts modality informativeness, cross-modal relation, and final action in a three-stage cascade.

Contributions. This paper makes three contributions. First, we formalize multimodal conflict handling as a five-category protocol with oracle actions and evaluate it on our 'Conflict Suite,' a balanced benchmark built from VQAv2 and MSCOCO. Second, we introduce CARM, a frozen-backbone cascade that separates modality informativeness, cross-modal relation, and action selection. Third, on this controlled Qwen2.5-VL-7B testbed and the canonical full split, we show that the resulting structure improves final decision quality: CARM reaches

66.2% task success on a 6,742-example test split, improving on the strongest non-learned baseline from 48.9% to 66.2%, and our ablations show that hidden-state inputs are the main source of the gain while the cascade itself raises task success from 65.1% to 66.2% over a flat hidden-state control.

2 Related Work

Conflict and hallucination in MLLMs. Modern MLLMs connect a pretrained vision encoder to a language model through a learned cross-modal interface (Li et al., 2023a; Liu et al., 2023; Dai et al., 2023). Although these systems perform well on aligned inputs, they remain brittle when image and text disagree, often hallucinating from the wrong modality or failing to ground their final answer (Li et al., 2023b; Favero et al., 2024; Seth et al., 2025). Recent surveys frame this as part of a broader multimodal hallucination and reliability problem, and related work also studies cross-modality knowledge conflict as a distinct failure source (Chen et al., 2025; Zhu et al., 2024). Our work focuses on this input-conflict failure mode directly.

Benchmarks and modality bias. Recent benchmarks make modality conflict measurable at scale. CLASH introduces controlled object- and attribute-level contradictions (Popordanoska et al., 2025), while MMIR extends inconsistency evaluation to layout-rich artifacts such as webpages, slides, and posters (Yan et al., 2025). Subsequent evaluations show that conflict degrades performance even when each modality is individually informative (Zhang et al., 2025c). Related work also documents strong modality bias and instance-dependent dominance in multimodal systems, including cases where models over-trust text under inconsistency (Hua et al., 2025; Zhang et al., 2025b; Jia et al., 2025; Deng et al., 2025). Mechanistic analyses further suggest that arbitration behavior is reflected in latent representations and specialized internal components rather than only in final outputs (Zhang et al., 2025a, 2026). Unlike work that only diagnoses the failure, we cast conflict resolution as a structured action-selection problem.

Abstention and selective prediction. Selective prediction trades coverage for risk by abstaining on examples where a model is likely to err (Geifman and El-Yaniv, 2017; Kamath et al., 2020; Wen et al., 2025). In VQA, reliable-answering work shows that simple confidence thresholding can achieve low risk only at very low coverage (Whitehead et al., 2022). Multimodal conflict requires a more structured decision: a system should abstain for different reasons in contradictory and both-weak cases, while still answering when exactly one modality is informative. CARM addresses this by predicting modality informativeness, the congruence or incongruence of the implied answers of each modality to the target question, and final action, explicitly rather than learning a single abstention threshold.

Table 1: Conflict protocol categories and oracle actions. Each category is defined by the cross-product of per-modality informativeness and pairwise consistency, with a deterministic oracle action.

Cat.	Vision info.?	Text info.?	Relation	Oracle action
C1	yes	yes	consistent	require agreement
C2	yes	no	—	trust vision
C3	no	yes	—	trust text
C4	yes	yes	contradictory	abstain
C5	no	no	—	abstain

3 Task Setting

Decision problem. Given image I, text caption T, and question q, the system predicts an action $\hat{a}$ and, when appropriate, an answer $\hat{y}$. We use four actions: `trust_vision`, `trust_text`, `require_agreement`, and `abstain`. A correct decision depends on two modality-specific properties (whether vision and text are informative for q) and one pairwise property (whether the informative modalities agree).

Five-category protocol. Table 1 summarizes the protocol we developed to evaluate these possible behaviors. Category 1 (C1) and Category 4 (C4) are scenarios where both modalities contained relevant information with which to answer the question; they differ only in whether those modalities are consistent over the question (C1) or inconsistent (C4). C2 and C3 are asymmetric single-modality cases, and C5 captures the case where the answer-relevant signal in both modalities is unclear.

Conflict Suite construction and evaluation. We build a "Conflict Suite" from VQAv2/COCO (Antol et al., 2015; Goyal et al., 2017) and assign proto-

col labels programmatically from modality-specific supervision fields derived with the Qwen2.5-VL-7B backbone. The resulting corpus contains 44,948 examples and is nearly balanced across C1–C5. Unless otherwise noted, we report results on the canonical prepared split: 31,463 training examples, 6,743 validation examples, and 6,742 test examples. We use VQAv2/COCO because it offers controlled answer supervision and reproducible conflict construction at scale; complementary reliability benchmarks such as POPE, CLASH, and MMIR target hallucination or contradiction detection, but not the full five-way arbitration decision studied here (Li et al., 2023b; Popordanoska et al., 2025; Yan et al., 2025). Appendix A reports the exact corpus and split counts used in the main comparison.

Dataset construction required systematic perturbation of the inconsistent modality. In the initial joined VQA + MSCOCO state, generally, both the caption and the image contained the information necessary to answer the question. This corresponded to a Category 1 type scenario in our taxonomy. To create Category 2 scenarios, where the caption did not answer the question (but otherwise was similar) we processed all C2 captions by passing them to an LLM-as-annotator (GPT-5-nano) and instructing the model to change the caption so it no longer answered the question (for example, removed the word "red" from the caption when the question asks for the color). Upon inspection, this was highly effective.

To build C3, we needed to change images so they no longer answered the question. To achieve this, we leveraged the fact that the COCO images have existing object annotations, for objects across the 80 COCO categories (e.g., [person, tennis racket]). We first limited the question-image-caption-answer scenarios to only those where the COCO annotated object appears verbatim in the question, so that it is likely the question is asking about the COCO object (i.e., ["tennis racket"] is in the question). Then, we embedded all images not yet included in the dataset, as well as the target image, using CLIP embeddings. Then, after excluding all images that also contained the target image's COCO objects (so none with a tennis racket) we selected the closest image in embedding space and swapped it in. Thus, it likely was similar to the original (close in embedding space) but lacked an object in the question, and thus could not be used to answer the question.

Category 4, which required contradicting implied answers from caption and image, used the text perturbation approach with GPT-5-nano, except asking the LLM to change the answer the question implied. There was no need to perturb the image as contradiction is symmetric, and there was no longer a "golden" answer at all.

Category 5 was constructed by swapping images as before, and perturbing text for irrelevance (not difference), as before. These dataset construction methods allowed us to cover the range of modality conflict scenarios that we hoped to test.

Metrics. Our primary metric is task success (TS): a prediction succeeds only if it takes the right arbitration action and, on answerable categories, returns the correct answer. This makes abstention correct for C4/C5 and correct answering necessary for C1–C3. We also report action accuracy, coverage (the fraction of examples on which the system answers), and accuracy-on-answered (AoA). Under this definition, the protocol oracle achieves TS $= 1.0$ by construction. Because that ceiling is definitionally built into the protocol, Appendix E also reports a practical fixed-output ceiling: under the same four protocol actions and frozen probe outputs, an oracle router would reach 89.5% task success on this full test split. CARM therefore captures 74.0% of the attainable task success under the operational ceiling induced by the frozen backbone outputs. Appendix A gives the exact task-success rule and representative cases, and Appendix D reports the full evaluation tables.

4 CARM

4.1 Overview

Arbitrating between conflicting modalities requires three kinds of evidence: a signal for whether vision is useful, a signal for whether text is useful, and a signal for whether the two modalities agree when both matter. CARM extracts these signals from a frozen Qwen2.5-VL-7B backbone and then maps them to an action with a small cascade of prediction heads. Figure 1 shows the full pipeline.

4.2 Features from a Frozen Backbone

Arbitration requires evidence about each modality separately and jointly. We therefore run three backbone passes: a vision-only probe, a text-only probe, and a full multimodal pass. The probe distributions are defined over a small family-specific VQA vocabulary selected by question type. In the reported Qwen2.5-VL-7B runs, this vocabulary is fixed by the backbone configuration rather than mined from

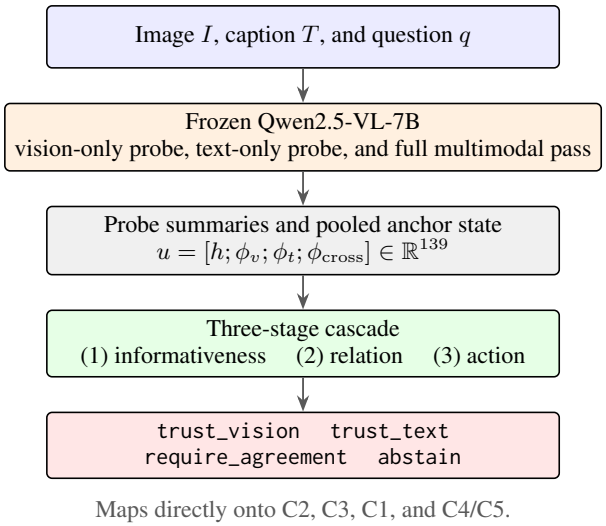

Figure 1: CARM architecture. CARM keeps the backbone frozen, extracts unimodal and cross-modal evidence, and makes a staged arbitration decision.

the training split: existence uses {yes, no, unknown}, count uses integers 0–20 plus `unknown`, and attribute-color uses 11 canonical color labels plus `unknown`. Appendix C.1 gives the exact ontology and canonicalization details. The full multimodal pass yields a fixed-width anchor-state sequence from the last hidden layer of the multimodal prompt. We summarize each probe with entropy, top-two margin, and max probability, add a small set of agreement features, and concatenate those summaries with the mean-pooled 128-dimensional anchor-state vector:

$$u = [h; \phi_v; \phi_t; \phi_{\text{cross}}] \in \mathbb{R}^{139}.$$

This combines lightweight confidence signals with a richer joint representation that is not available from output probabilities alone. Appendix C gives the exact formulas and prompt templates.

4.3 Cascade Arbitration

A flat action classifier would conflate decisions with different prerequisites. Choosing `trust_vision` or `trust_text` requires knowing which modality is informative, while abstaining in C4 requires knowing that both modalities are informative and contradictory. CARM mirrors that logic with a three-stage cascade over a shared 128-dimensional trunk. Stage 1 predicts vision and text informativeness, Stage 2 predicts the cross-modal relation (consistent, contradictory, asymmetric, or both-weak), and Stage 3 predicts the final action conditioned on the earlier stages. In asymmetric cases, the final head uses the Stage 1 informativeness predictions to choose between `trust_vision` and `trust_text`. We train all stages jointly while keeping the backbone strictly frozen. This keeps the decision process auditable and adds only 36,236 trainable parameters relative to the 7B backbone.

4.4 Training Objective

We train CARM against the explicit arbitration structure rather than only the final action. Each example provides supervision for vision informativeness, text informativeness, pairwise relation, and final action. For Cascade CARM, we optimize the weighted multi-task objective:

$$\mathcal{L} = \lambda_v \, \mathrm{CE}(\hat{l}_v, l_v) + \lambda_t \, \mathrm{CE}(\hat{l}_t, l_t) + \lambda_r \, \mathrm{CE}(\hat{l}_{\text{rel}}, l_{\text{rel}}) + \lambda_a \, \mathrm{CE}(\hat{l}_{\text{act}}, l_{\text{act}}).$$

This objective encourages the model to learn the intermediate arbitration decisions that lead to the final action, instead of collapsing the whole problem into a single flat label. The distribution variants use the same staged objective with distribution inputs in place of hidden-state inputs, while Flat Hidden keeps only the final action loss and removes the auxiliary heads. Exact loss weights and optimization settings are given in Section 5 and Appendix C.

5 Experimental Setup

We evaluate on the canonical full split of Conflict Suite (31,463 training, 6,743 validation, and 6,742 test examples). Appendix A reports the exact corpus counts, split construction, and label derivation.

All experiments use frozen Qwen2.5-VL-7B in bfloat16 on a single NVIDIA A100 40 GB GPU. To keep backbone access comparable, every probe-based baseline and every learned variant uses the same three backbone queries: a vision-only probe, a text-only probe, and a full multimodal pass. Appendix C gives the exact prompt wrappers, masking rules, and CARM feature definitions.

We compare three method families. The first is a set of non-learned baselines: backbone direct (full multimodal decoding without abstention), agreement check (answer only when unimodal probes agree), confidence threshold (validation-tuned abstention on backbone confidence), probe heuristic (entropy-based routing between the two unimodal probes), and prompt-only abstain (zero-shot prompting over a closed action set). We keep probe heuristic because it is the simplest non-learned

router that can explicitly choose between modalities rather than only answer or abstain. The second family is a pair of distribution-based learned variants (Dist CARM v1 and Dist CARM v2) that use multimodal, vision-only, and text-only answer distributions plus cross-modal features, but no hidden-state input. The third is a hidden-state comparison between Flat Hidden, which predicts the final action in one step, and Cascade CARM, which uses the same hidden-state input but decomposes the problem into informativeness, relation, and action.

This comparison suite is designed to answer a small set of identification questions rather than to maximize architectural breadth. The non-learned baselines test whether conflict handling can be explained by generic abstention or agreement heuristics alone. The distribution-based variants test whether the relevant arbitration signal is already present in modality-conditional answer distributions, without appealing to internal hidden states. The Flat Hidden versus Cascade comparison then isolates the architectural contribution of the cascade itself under matched hidden-state access. In other words, the empirical design is intended to establish not only that CARM works, but also why it improves over simpler alternatives.

The training objective is described in Section 4.4. At a high level, all learned variants keep the backbone frozen, optimize with AdamW, and early-stop on validation task success. Cascade CARM and Dist CARM v1 use the same staged multi-task objective and a 10-epoch budget, Dist CARM v2 keeps that objective but extends the training budget and upweights `trust_text`, and Flat Hidden drops the auxiliary supervision and trains only the final action head. Appendix B and Appendix C give the exact per-method settings and implementation details. The full-data Dist CARM v2 rerun hit the Quest 48-hour wall-time before final test evaluation, so the main comparison reports only completed full-data test runs and treats the class-weighted variant as incomplete evidence.

6 Results

6.1 Main Results

Table 2 summarizes the main comparison on the 6,742-example full test split. Cascade CARM reaches 66.2% task success, outperforming the strongest non-learned baseline, agreement check, by 17.3 points (48.9% $\rightarrow$ 66.2%). The exact Wilson interval for Cascade is $[65.1, 67.3]$, and the paired comparison against agreement check is decisively significant under a continuity-corrected McNemar test (Appendix E).

Table 2: Main results on the 6,742-example full test split. Higher is better. Dist CARM v2 (+wt) is omitted because its full-data rerun hit the 48-hour cluster wall-time before final test evaluation.

Method	TS ↑	Cov ↑	AoA ↑	Act. acc. ↑
Backbone direct	.438	**1.000**	.601	—
Agreement check	.489	.542	.631	—
Conf. threshold	.470	.680	.638	—
Probe heuristic	.428	.956	.565	—
Prompt-only abstain	.473	.465	.742	.389
Dist CARM v1	.588	.389	**.791**	.509
Flat Hidden	.651	.543	.758	**.666**
Cascade CARM	**.662**	.589	.772	.647

Note. Higher is better. Rows without an abstention mechanism have coverage 1.000 by definition.

The baseline pattern highlights why structured arbitration matters. Backbone direct and confidence threshold answer almost everything, but they fail whenever abstention is the correct action. Agreement check is safer on blatant contradictions, yet it abstains too often when disagreement merely indicates that one modality is irrelevant. CARM is the only completed method that simultaneously keeps high answered accuracy (77.2% AoA) and enough coverage (58.9%) to improve end-to-end task success on the full split. Relative to the fixed-output oracle ceiling of 89.5% (Appendix E), CARM captures 74.0% of the attainable task success under the current action space and frozen backbone outputs.

Most of the gain comes from avoiding catastrophic failure on the asymmetric routing cases. Table 3 shows that CARM improves C1–C3 substantially over the distribution-only variants while remaining competitive on C4 and C5. The main remaining weakness is C3, where some text-informative examples still resemble both-weak cases under simple probe summaries.

6.2 Ablations

Table 3 isolates which parts of CARM matter. Hidden-state input is the main driver on the same

Table 3: Ablation results on the 6,742-example full test split by protocol category. Dist v1 uses answer distributions and cross-modal features only; Flat Hidden and Cascade use the same 139-dimensional anchor-state-plus-probe input. Dist CARM v2 (+wt) is omitted because its full-data rerun timed out before final test evaluation.

Variant	TS↑	C1↑	C2↑	C3↑	C4↑	C5↑
Dist v1	.588	.619	.501	.213	.758	**.850**
Flat Hidden	.651	.678	.678	.419	**.764**	.717
Cascade	**.662**	**.734**	**.736**	**.456**	.683	.702

full split: replacing distribution inputs with hidden-state features improves task success from 0.588 under Dist CARM v1 to 0.651 under Flat Hidden. The cascade then adds a smaller but still positive improvement, raising task success from 0.651 to 0.662 on the same hidden-state input family.

This ablation structure follows the claim logic of the paper. If the distribution-only models matched Cascade, then the main benefit would be attributable to output-space arbitration rather than hidden-state representations. If Flat Hidden matched Cascade, then the hidden-state representation alone would explain the gain and the staged decomposition would add little. The observed ordering therefore matters substantively: it indicates that hidden-state access is necessary for strong performance, but that the hierarchical decomposition still contributes additional decision quality beyond a one-step action head.

The gains are concentrated in the asymmetric routing categories. C2 rises from 0.501 under Dist v1 to 0.736 under Cascade, and C3 rises from 0.213 to 0.456. Flat Hidden already improves strongly over the distribution model on the same action space, but Cascade still adds a further boost on answerable categories, especially C2 (0.678 $\rightarrow$ 0.736) and C3 (0.419 $\rightarrow$ 0.456), by conditioning the final action on informativeness and relation predictions. The distribution and flat models remain stronger on C4/C5 because they abstain more conservatively, but that operating point gives up too much answerable mass overall (Table 3).

7 Conclusion

We presented CARM, a lightweight arbitration layer (36,236 trainable parameters) for multimodal conflict handling. Instead of treating disagreement as generic uncertainty, CARM decomposes the problem into modality informativeness, cross-modal relation, and action selection. On Conflict Suite with Qwen2.5-VL-7B, this structure improves task success from 48.9% to 66.2% over the strongest non-learned baseline, and our full-data ablations show that hidden-state features are the dominant signal while the cascade adds a smaller but consistent additional gain over a flat hidden-state control. The main remaining challenge is separating text-informative cases from both-weak cases, which suggests that future progress will require richer cross-modal representations rather than more aggressive abstention alone.

Limitations

Single-backbone evidence. Our main empirical claim is about what works for Qwen2.5-VL-7B on this benchmark. We do not show that the same hidden-state signal or cascade architecture will transfer unchanged to other MLLMs with different fusion mechanisms.

Scope of evaluation. The benchmark uses English-language factual questions over everyday photographs derived from VQAv2/COCO. We do not evaluate on naturally occurring multimodal conflicts in domains such as medical imaging, document understanding, or chart QA.

Limited stability evidence. The headline result comes from one completed full-data run per method on one canonical split. We do not yet have same-split multi-seed estimates for the full-data setting, and the stronger class-weighted Dist CARM v2 full-data comparator timed out under the 48-hour cluster limit before final test evaluation. Stronger stability claims would require additional same-split cascade seeds, a completed rerun of the class-weighted distribution baseline, and preferably one additional backbone.

References

Stanislaw Antol, Aishwarya Agrawal, Jiasen Lu, Margaret Mitchell, Dhruv Batra, C. Lawrence Zitnick, and Devi Parikh. 2015. VQA: Visual question answering. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*.

Zhiyuan Chen, Yuecong Min, Jie Zhang, Bei Yan, Jiahao Wang, Xiaozhen Wang, and Shiguang Shan. 2025. A Survey of Multimodal Hallucination Evaluation and Detection.

Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. 2023. InstructBLIP: Towards General-purpose Vision-Language Models with Instruction Tuning. *arXiv preprint*. ArXiv:2305.06500 [cs].

Ailin Deng, Tri Cao, Zhirui Chen, and Bryan Hooi. 2025. Words or Vision: Do Vision-Language Models Have Blind Faith in Text?

Alessandro Favero, Luca Zancato, Matthew Trager, Siddharth Choudhary, Pramuditha Perera, Alessandro Achille, Ashwin Swaminathan, and Stefano Soatto. 2024. Multi-Modal Hallucination Control by Visual Information Grounding. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14303–14312, Seattle, WA, USA. IEEE.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. *arXiv preprint arXiv:1705.08500*.

Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2017. Making the V in VQA matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Tianze Hua, Tian Yun, and Ellie Pavlick. 2025. How Do Vision-Language Models Process Conflicting Information Across Modalities? *arXiv preprint*. ArXiv:2507.01790 [cs] version: 1.

Yanhao Jia, Ji Xie, S. Jivaganesh, Hao Li, Xu Wu, and Mengmi Zhang. 2025. Seeing Sound, Hearing Sight: Uncovering Modality Bias and Conflict of AI models in Sound Localization. *arXiv preprint*. ArXiv:2505.11217 [cs].

Amita Kamath, Robin Jia, and Percy Liang. 2020. Selective question answering under domain shift. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 5684–5696.

Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023a. BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. *arXiv preprint*. ArXiv:2301.12597 [cs].

Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. 2023b. Evaluating Object Hallucination in Large Vision-Language Models. *arXiv preprint*. ArXiv:2305.10355 [cs].

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023. Visual Instruction Tuning. *arXiv preprint*. ArXiv:2304.08485 [cs].

Teodora Popordanoska, Jiameng Li, and Matthew B. Blaschko. 2025. CLASH: A Benchmark for Cross-Modal Contradiction Detection.

Ashish Seth, Dinesh Manocha, and Chirag Agarwal. 2025. Towards a Systematic Evaluation of Hallucinations in Large-Vision Language Models. *arXiv preprint*. ArXiv:2412.20622 [cs].

Bingbing Wen, Jihan Yao, Shangbin Feng, Chenjun Xu, Yulia Tsvetkov, Bill Howe, and Lucy Lu Wang. 2025. Know Your Limits: A Survey of Abstention in Large Language Models. *Transactions of the Association for Computational Linguistics*, 13:529–556.

Spencer Whitehead, Suzanne Petryk, Vedaad Shakib, Joseph Gonzalez, Trevor Darrell, Anna Rohrbach, and Marcus Rohrbach. 2022. Reliable visual question answering: Abstain rather than answer incorrectly. In *Proceedings of the European Conference on Computer Vision (ECCV)*.

Qianqi Yan, Yue Fan, Hongquan Li, Shan Jiang, Yang Zhao, Xinze Guan, Ching-Chen Kuo, and Xin Eric Wang. 2025. Multimodal Inconsistency Reasoning (MMIR): A New Benchmark for Multimodal Reasoning Models. *arXiv preprint*. ArXiv:2502.16033 [cs].

Yu Zhang, Jinlong Ma, Yongshuai Hou, Xuefeng Bai, Kehai Chen, Yang Xiang, Jun Yu, and Min Zhang. 2025a. Evaluating and Steering Modality Preferences in Multimodal Large Language Model.

Yu Zhang, Mufan Xu, Xuefeng Bai, Kehai chen, Pengfei Zhang, Yang Xiang, and Min Zhang. 2026. Instruction Anchors: Dissecting the Causal Dynamics of Modality Arbitration. *arXiv preprint*. ArXiv:2602.03677 [cs].

Zhuoran Zhang, Tengyue Wang, Xilin Gong, Yang Shi, Haotian Wang, Di Wang, and Lijie Hu. 2025b. When Modalities Conflict: How Unimodal Reasoning Uncertainty Governs Preference Dynamics in MLLMs. *arXiv preprint*. Version Number: 1.

Zongmeng Zhang, Wengang Zhou, Jie Zhao, and Houqiang Li. 2025c. Robust Multimodal Large Language Models Against Modality Conflict. *arXiv preprint*. ArXiv:2507.07151 [cs] version: 1.

Tinghui Zhu, Qin Liu, Fei Wang, Zhengzhong Tu, and Muhao Chen. 2024. Unraveling Cross-Modality Knowledge Conflicts in Large Vision-Language Models. *arXiv preprint*. ArXiv:2410.03659 [cs].

Appendix

A Conflict Suite Data Construction and Statistics

Label derivation. Each prepared example stores modality-specific supervision fields derived during dataset preparation, including whether vision and text support the target answer, supported targets when available, and a pairwise relation label. We map those fields deterministically into the protocol in Table 1. C1 corresponds to two informative, consistent modalities; C2 and C3 correspond to asymmetric single-modality support; C4 corresponds to two informative but contradictory modalities and always maps to `abstain`; and C5 corresponds to both modalities being uninformative and also maps to `abstain`. A Table A1 summarizes the full corpus and the canonical train/val/test split used in the main comparison in Table 2. The split preserves the near-balance of the full dataset while remaining almost uniform across the three answer families within each protocol category.

Table A1: Conflict Suite statistics for the canonical full-data split used in the paper's main comparison.

Split	Total	C1	C2	C3	C4	C5
Train	31,463	6,299	6,299	6,299	6,263	6,303
Val	6,743	1,350	1,350	1,350	1,342	1,351
Test	6,742	1,350	1,350	1,350	1,342	1,350

A.1 Representative Examples

The five representative cases in Table A3 correspond to concrete test examples from the prepared corpus. We show one image, caption, question, gold answer, and oracle action for each protocol category.

C1 example

Caption: A white bathroom sink sitting next to a white bathtub.
Question: What color is the sink?
Gold answer: white
Oracle action: `require_agreement`.

C2 example

Caption: A chair that has a thing on it.
Question: Is this an older style suitcase?
Gold answer: yes
Oracle action: `trust_vision`.

C3 example

Caption: A black and brown cat sitting in chair inside blue bag.
Question: What color is the backpack?
Gold answer: blue
Oracle action: `trust_text`.

C4 example

Caption: There is a bed with a green comforter and a metal headboard and foot board.
Question: What color is this beds comforter?
Gold answer: blue
Oracle action: `abstain`.

C5 example

Caption: A bird sitting on someone's arm in a room.
Question: What color is the bird?
Gold answer: blue
Oracle action: `abstain`.

A.2 Protocol Outcome Rules and Representative Cases

Task success definition. In the revised protocol, task success is category-aware rather than tied only to the action name. For C1–C3, a system succeeds if it answers and the returned answer is correct. For C4 and C5, a system succeeds if it abstains. This is the exact logic implemented in the evaluation code.

Table A2: Task-success rule by category. In C1–C3, success requires a correct answer; in C4–C5, success requires abstention.

Category	Oracle action	TS is 1 if...
C1	require agreement	answers and answer is correct
C2	trust vision	answers and answer is correct
C3	trust text	answers and answer is correct
C4	abstain	abstains
C5	abstain	abstains

Representative cases. Table A3 shows one successful example for each protocol category. The examples make clear why the benchmark is an arbitration problem: disagreement can mean contradiction (C4), harmless irrelevance (C2/C3), or genuine agreement (C1).

Table A3: Representative test examples under the revised protocol. Captions are the text seen by the model.

Cat.	Caption seen by model	Question	Vision probe	Text probe	Successful outcome
C1	A white bathroom sink sitting next to a white bathtub.	What color is the sink?	white	white	require agreement, answer "white"
C2	A chair that has a thing on it.	Is this an older style suitcase?	yes	no	trust vision, answer "yes"
C3	A black and brown cat sitting in chair inside blue bag.	What color is the backpack?	pink	blue	trust text, answer "blue"
C4	There is a bed with a green comforter and a metal headboard and foot board.	What color is this beds comforter?	blue	green	abstain because both modalities are informative but contradictory
C5	A bird sitting on someone's arm in a room.	What color is the bird?	brown	blue	abstain because neither modality is reliable enough

B Method-Specific Setup

Tables B1 and B2 summarize the exact setup used for each method in Section 5. The main comparison keeps the dataset split fixed and changes only how each method turns frozen backbone outputs into an arbitration action.

Table B1: Method-specific setup for non-learned baselines in the main comparison.

Method	Backbone evidence	Decision rule	Additional details
Backbone direct	full multimodal pass	decode an answer directly; never abstain	no learned arbitration module
Agreement check	vision and text probes	answer only when unimodal top-1 answers match; else abstain	no tuning or training
Conf. threshold	full multimodal pass	answer only when backbone confidence exceeds a validation-tuned threshold	single-threshold selective prediction baseline
Probe heuristic	vision and text probes	choose the lower-entropy probe unless both probe entropies exceed a validation-tuned threshold	strongest simple non-learned router
Prompt-only abstain	free-form action prompt plus vision and text probes	generate one action label, then execute that action with the two unimodal probes	zero-shot; no learned parameters

Table B2: Method-specific setup for learned variants in the main comparison.

Method	Backbone evidence	Decision rule	Additional details
Dist CARM v1	multimodal, vision-only, and text-only answer distributions plus cross-modal features	learned three-stage cascade without hidden-state input	110-dimensional distribution input; early stop on validation TS
Dist CARM v2 (+wt)	same as Dist CARM v1	same cascade as Dist CARM v1	trust_text upweighted $3\times$ via action class weights $[1, 3, 1, 1]$; full-data rerun timed out at the 48-hour cluster limit before test evaluation (best val TS .615)
Flat Hidden	pooled anchor states plus probe and cross-modal features	learned single-step action head	same 139-dimensional input as Cascade CARM, but no staged decomposition
Cascade CARM	pooled anchor states plus probe and cross-modal features	learned three-stage cascade over informativeness, relation, and action	36,236 trainable parameters; early stop on validation TS

Relative to the non-learned baselines, the learned variants all use the same frozen backbone calls. Dist CARM v1 tests whether answer distributions alone are enough, Dist CARM v2 repeats that comparison with class weighting but is incomplete on the full split because of the cluster wall-time cap, Flat Hidden adds the anchor-state representation but predicts the action in one step, and Cascade CARM uses the same input as Flat Hidden while imposing the staged arbitration structure.

C CARM Prompts and Implementation Details

C.1 Answer Vocabulary and Canonicalization

The answer vocabulary is family-specific. In the reported Qwen2.5-VL-7B runs, we use a fixed ontology rather than a vocabulary mined from the training split: existence uses {yes, no, unknown}, count uses integers 0–20 plus `unknown`, and attribute-color uses 11 canonical color labels (`red`, `blue`, `green`, `yellow`, `black`, `white`, `brown`, `gray`, `orange`, `pink`, `purple`) plus `unknown`. The question family determines which of these vocabularies is active for the vision-only, text-only, and multimodal queries. The implementation also supports optional family-specific overrides derived from the training data, but we do not use that path in the reported experiments.

Generated strings are canonicalized back into the same family ontology before agreement checks and evaluation. Existence accepts aliases such as `y`/`true`/`1` and `n`/`false`/`0`; count accepts digits or spelled-out numbers zero through twenty; and color normalization maps aliases such as `grey` $\rightarrow$ `gray` and `violet` $\rightarrow$ `purple`.

Backbone QA prompt. All backbone queries share the same wrapper prompt, with the second line determined by the answer family:

```
You are answering a VQA question.
[family-specific instruction]
Caption: {caption}
Question: {question}
Answer:
```

For existence questions, the instruction is "Answer yes or no only." For count questions, it is "Answer with a single integer only." For attribute-color questions, it is "Answer with a single color word only." The multimodal pass uses both image and caption, the vision-only probe keeps the image but omits the caption line, and the text-only probe uses the caption and question without an image.

Prompt-only abstain baseline. The prompt-only baseline directly predicts one of the four arbitration actions with the following template:

```
You are a multimodal arbitration system.
Given the image, caption, and question, return exactly one label from this
closed set:
TRUST_VISION
TRUST_TEXT
REQUIRE_AGREEMENT
ABSTAIN
Do not explain.
Caption: {caption}
Question: {question}
Decision:
```

Backbone queries. The vision-only and text-only probes produce answer distributions at the first generated token:

$$p_v(y \mid I, q) = \mathrm{softmax}(\mathrm{Backbone}(I, \varnothing, q))_y,$$
$$p_t(y \mid T, q) = \mathrm{softmax}(\mathrm{Backbone}(\varnothing, T, q))_y.$$

Masking is implemented at the input level. For the vision-only probe, the model receives the image and question but no caption text. For the text-only probe, the model receives the caption and question with no image.

Hidden-state feature. The full multimodal pass yields a fixed-width anchor-state sequence $H_{\text{anchor}} \in \mathbb{R}^{32\times128}$ from the last hidden layer: the backbone adapter truncates or pads the raw hidden states to a common sequence length and hidden size before the CARM heads see them. CARM then mean-pools that sequence:

$$h = \text{mean-pool}(H_{\text{anchor}}) \in \mathbb{R}^{128}.$$

There is no additional learned projection inside the experimental heads.

Probe summaries and agreement features. For each probe distribution we compute entropy, top-two margin, and max probability:

$$\phi_v = [\mathrm{ent}(p_v),\ \mathrm{mar}(p_v),\ \max(p_v)] \in \mathbb{R}^3,$$
$$\phi_t \in \mathbb{R}^3 \text{ (analogously).}$$

To capture cross-modal agreement, we define five additional features:

1. binary agreement $a_{\text{bin}} = \mathbf{1}[\arg\max p_v = \arg\max p_t]$,
2. soft overlap $\sum_y \min(p_v(y), p_t(y))$,
3. $\max(p_v)$,
4. $\max(p_t)$,
5. `conf_disagree` $= \max(p_v)\max(p_t)(1 - a_{\text{bin}})$.

These form $\phi_{\text{cross}} \in \mathbb{R}^5$, and the final CARM input is $u = [h; \phi_v; \phi_t; \phi_{\text{cross}}] \in \mathbb{R}^{139}$.

Stage 2 relation labels. Stage 2 training targets are derived from the protocol category: C1 examples receive label consistent, C4 examples receive contradictory, C2 and C3 both receive asymmetric, and C5 receives both_weak. This gives a four-class problem that matches $\hat{l}_{\text{rel}} \in \mathbb{R}^4$. The final action head then combines this relation prediction with the Stage 1 informativeness outputs to distinguish C2 (`trust_vision`) from C3 (`trust_text`).

Cascade parameterization. The shared trunk is

$$z = \text{GELU}(W_{\text{trunk}} u + b) \in \mathbb{R}^{128}.$$

Stage 1 predicts vision and text informativeness:

$$\hat{l}_v = W_{v\text{-info}} z \in \mathbb{R}^2, \qquad \hat{l}_t = W_{t\text{-info}} z \in \mathbb{R}^2.$$

Stage 2 predicts relation classes conditioned on Stage 1:

$$\begin{aligned} z_{\text{rel}} &= \text{GELU}(W_{\text{rel1}}[z; \sigma(\hat{l}_v); \sigma(\hat{l}_t)]), \\ \hat{l}_{\text{rel}} &= W_{\text{rel2}} z_{\text{rel}} \in \mathbb{R}^4. \end{aligned}$$

Stage 3 predicts the final action conditioned on both earlier stages:

$$\begin{aligned} z_{\text{act}} &= \text{GELU}(W_{\text{act1}}[z; \sigma(\hat{l}_v); \sigma(\hat{l}_t); \sigma(\hat{l}_{\text{rel}})]), \\ \hat{l}_{\text{act}} &= W_{\text{act2}} z_{\text{act}} \in \mathbb{R}^4. \end{aligned}$$

Training. Cascade CARM is trained with AdamW (learning rate 10^{-3}, weight decay 10^{-2}, batch size 1) and early stopping on validation task success, with a maximum of 10 epochs and patience 3. The action loss is weighted most heavily, with $\lambda_{\text{vision}} = 1.0$, $\lambda_{\text{text}} = 1.0$, $\lambda_{\text{relation}} = 1.5$, and $\lambda_{\text{action}} = 3.0$. On the main run, validation task success peaks at epoch 1 (0.663), and early stopping triggers after epoch 4. The Flat Hidden control uses the same backbone features but removes the auxiliary heads, peaks at epoch 3 (0.656), and stops after epoch 6.

Inference. At inference time, `trust_vision` answers from (I, q) with the text suppressed; `trust_text` answers from (T, q) with the image masked; `require_agreement` returns the shared unimodal answer when the probes agree and abstains otherwise; and `abstain` emits a standardized refusal.

D Evaluation Details

Table D1 reports the exact test metrics used in the main comparison, and Table D2 breaks task success out by protocol category for the learned variants.

Table D1: Exact full-data test metrics with Wilson 95% confidence intervals for task success. Dist CARM v2 (+wt) is omitted because its full-data rerun timed out before final test evaluation.

Method	Cov.↑	AoA↑	TS↑	95% CI	ActAcc↑	Δ TS vs. agreement
Backbone direct	**1.000**	.601	.438	[.426, .450]	—	−.051
Agreement check	.542	.631	.489	[.478, .501]	—	—
Conf. threshold	.680	.638	.470	[.459, .482]	—	−.019
Probe heuristic	.956	.565	.428	[.416, .440]	—	−.061
Prompt-only abstain	.465	.742	.473	[.461, .485]	.389	−.016
Dist CARM v1	.389	**.791**	.588	[.576, .599]	.509	+.098
Flat Hidden	.543	.758	.651	[.639, .662]	**.666**	+.161
Cascade CARM	.589	.772	**.662**	[.651, .673]	.647	+.173

Table D2: Per-category task success for completed full-data learned variants. Dist CARM v2 (+wt) is omitted because its full-data rerun timed out before test evaluation.

Variant	C1	C2	C3	C4	C5
Dist v1	.619	.501	.213	.758	**.850**
Flat Hidden	.678	.678	.419	**.764**	.717
Cascade	**.734**	**.736**	**.456**	.683	.702

E Additional Diagnostics

Practical ceiling under fixed outputs. The protocol oracle TS $= 1.0$ is definitionally correct but not operationally informative, because it assumes perfect answers once the right action is known. A more practical ceiling keeps all backbone outputs fixed and grants an oracle only the choice among the four protocol actions. Using the saved vision/text probe outputs and the standard execution semantics of `trust_vision`, `trust_text`, `require_agreement`, and `abstain`, this fixed-output oracle reaches 89.5% task success (6037/6742; 95% Wilson CI $[88.8, 90.3]$). By category, it reaches 90.3% on C1, 86.9% on C2, 70.6% on C3, and 100% on C4/C5. CARM therefore achieves 74.0% of the attainable task success under the current action space and frozen outputs, with most of the remaining gap concentrated in C3-like cases where the unimodal answers themselves are already unreliable.

Statistical comparison. Against agreement check, CARM improves task success from 48.9% to 66.2%. A paired McNemar test with continuity correction gives $\chi^2 = 609.5$ on the 6,742 paired test examples, strongly rejecting equal performance ($p \ll 10^{-16}$).

Why C3 remains hard. The main error pattern is C3/C5 confusability. On the test set, the vision-only probe is not uniformly weak in C3: $\max(p_v)$ averages 0.922 ± 0.157 in C3 versus 0.923 ± 0.153 in C5. The more informative signal is textual confidence: $\max(p_t)$ averages 0.905 ± 0.152 in C3 versus 0.828 ± 0.198 in C5. However, those distributions still overlap substantially, so the current scalar agreement features do not cleanly separate "text informative, vision weak" from "both weak." As a result, CARM is precise when it predicts `trust_text`, but it misses many valid C3 cases by defaulting to abstention.